

The good mic helped a bit

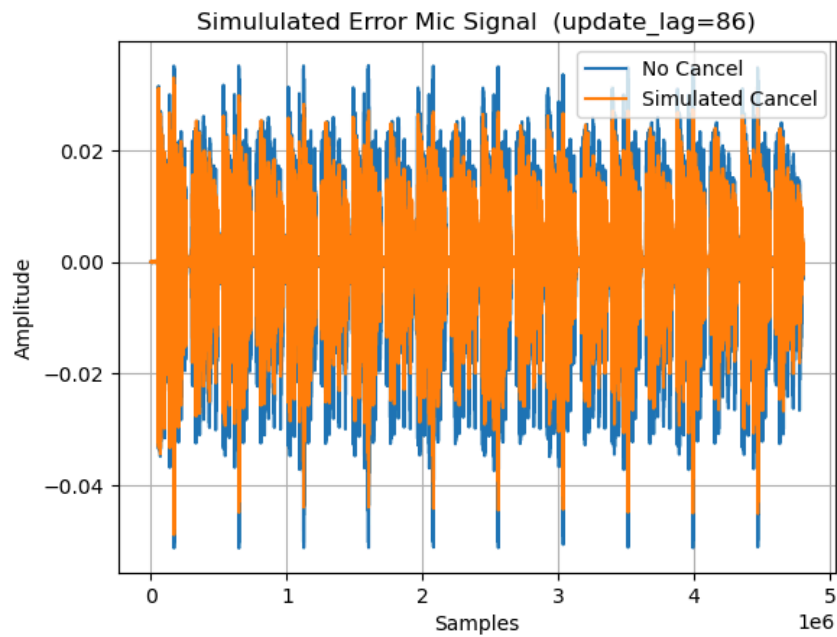
- Original error: sm58
- Original ref: pg58

	Played from source speaker					
Band	Source	Error pg58	Error sm58	Ref pg58	Ref sm58	Good mic
0-200 Hz	3.91	3.72	1.93	11.79	15.37	11.89
200-300 Hz	16.22	15.31	13.05	30.42	33.78	26.31
300-700 Hz	67.62	59.08	62.20	44.71	39.21	50.57
700-1k Hz	3.75	6.99	7.80	3.20	2.56	2.98
1k-2k Hz	6.80	8.49	8.10	6.25	5.43	5.60
2k-5k Hz	1.70	6.40	6.93	3.62	3.65	2.65

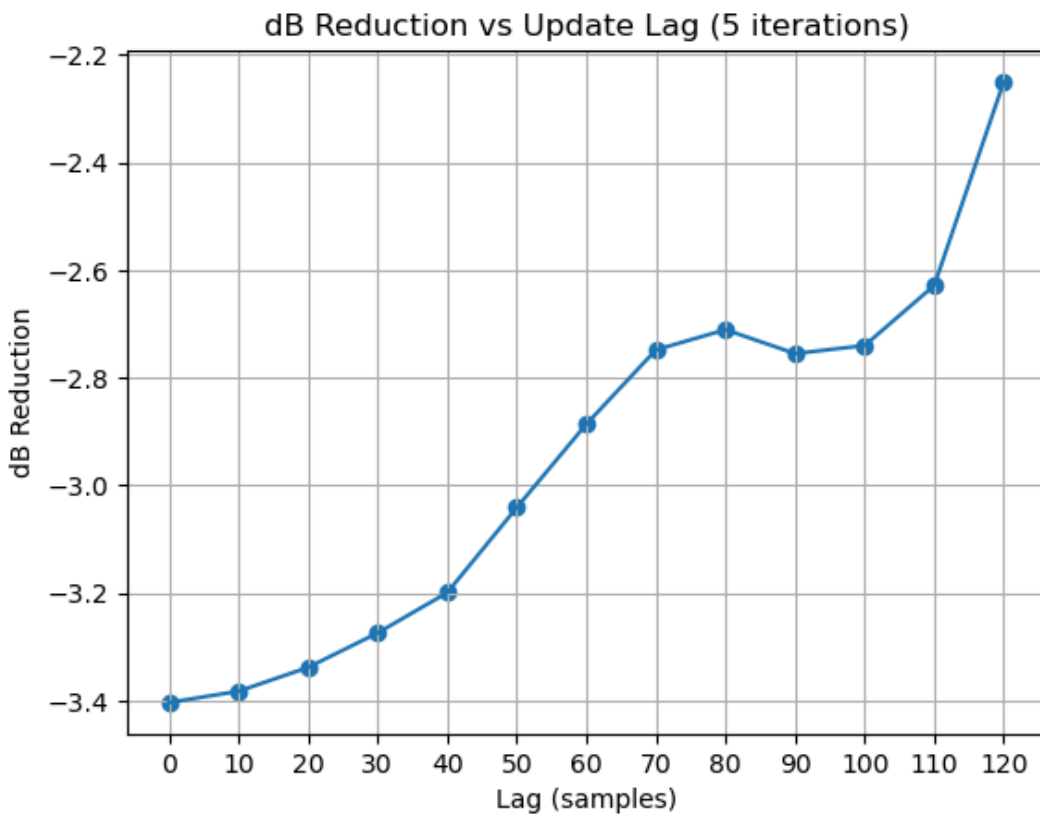
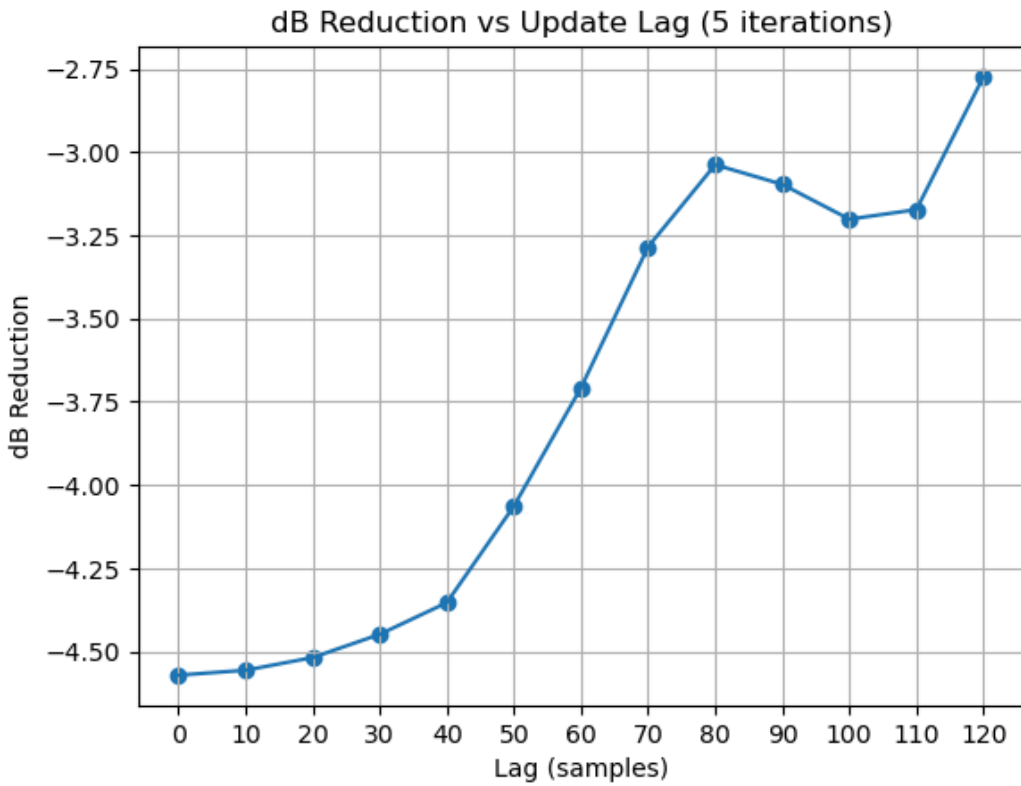
Lag cancellation is less stable but possible

- Smaller step size, more leak

-3.0 dB

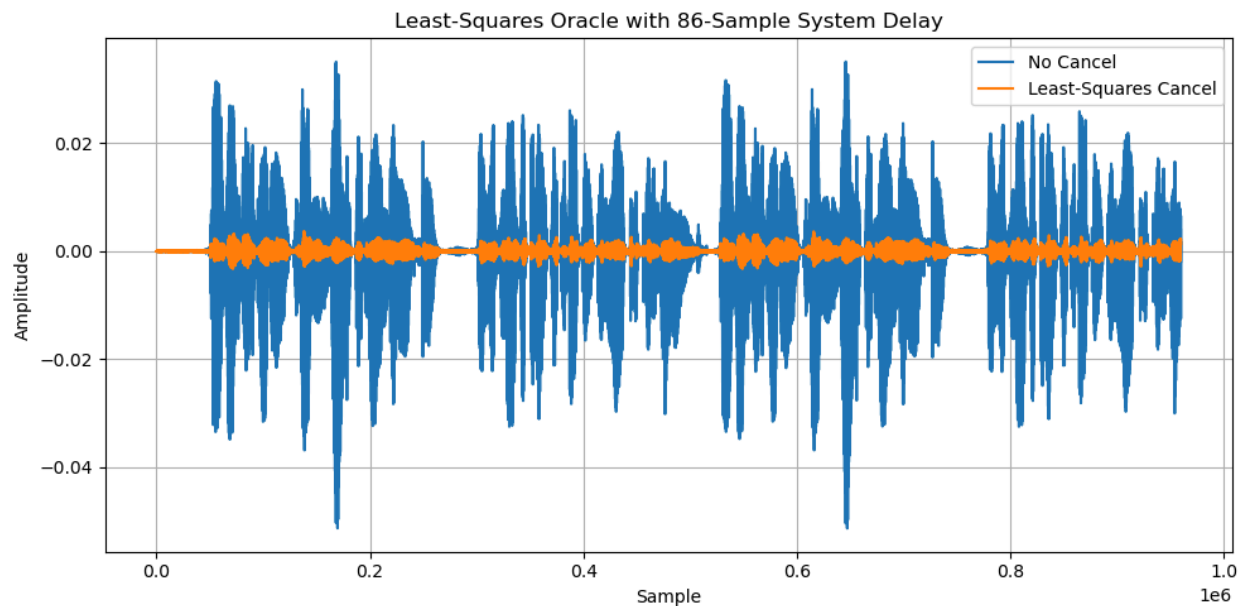


Lag vs performance curves

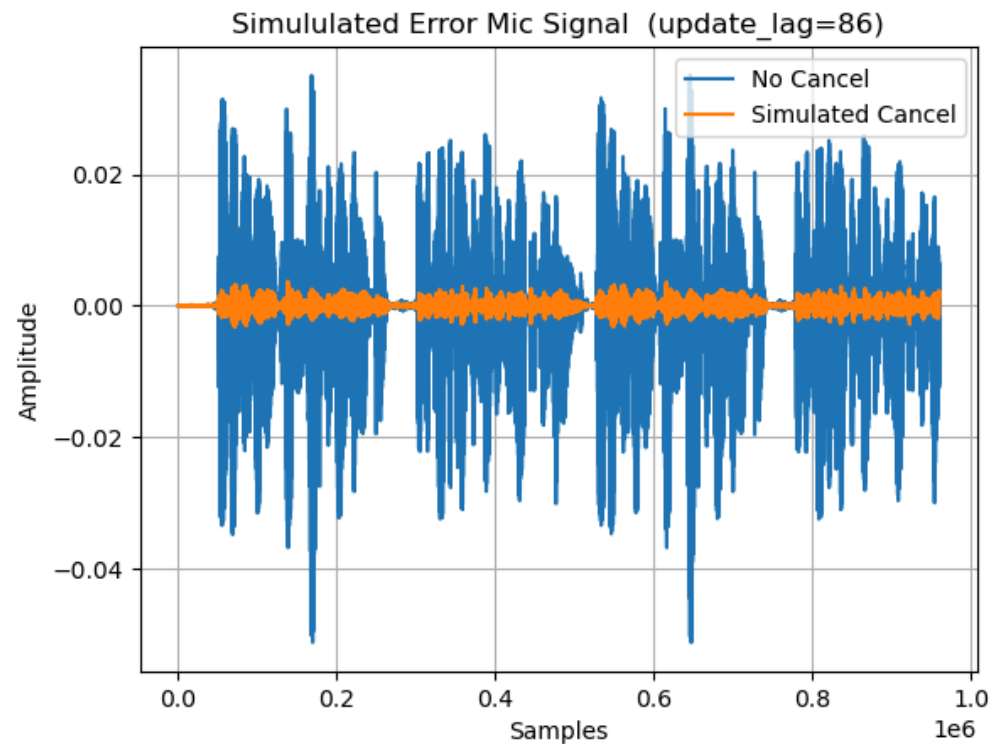


The lagged optimal filter works when frozen

Pure offline least squares -19.46dB

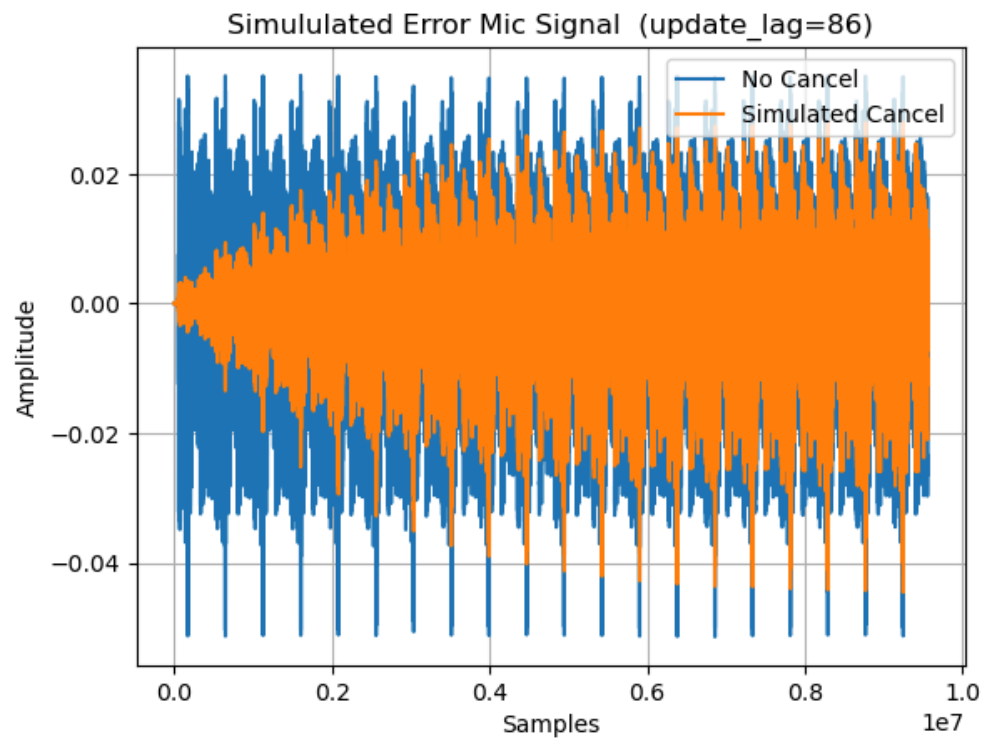


Frozen in FxLMS sample by sample -19.46dB

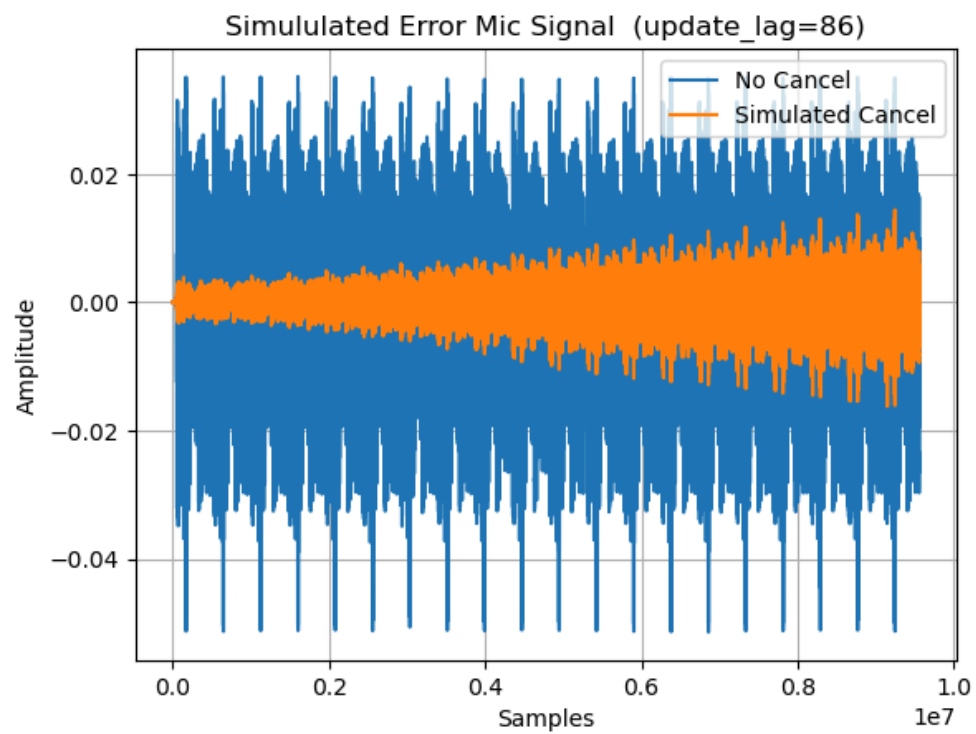


Learning from the filter destroyed it

With leak

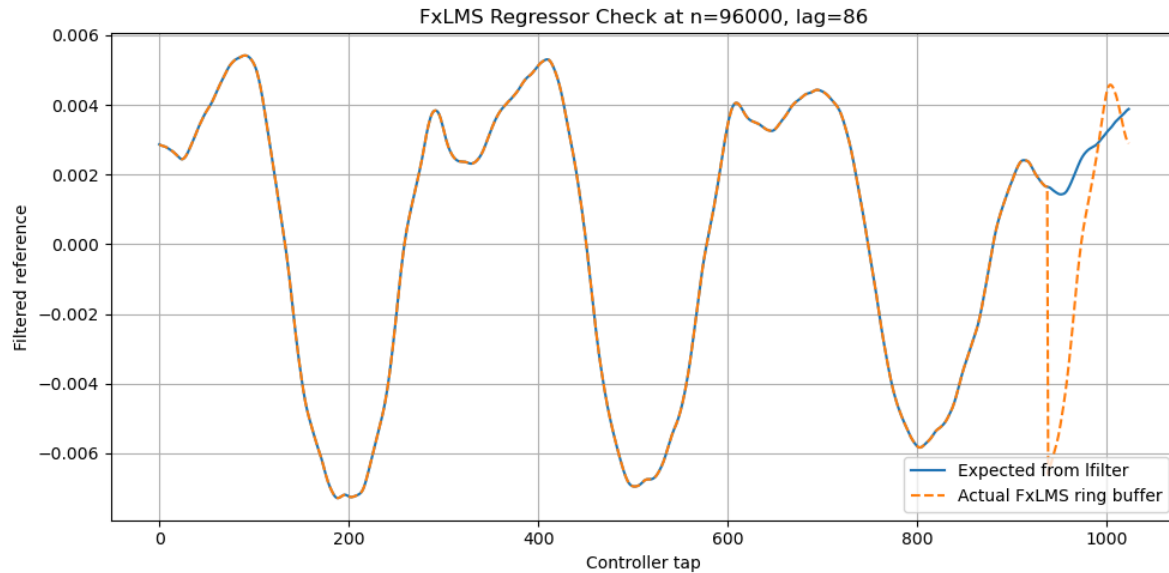


No leak

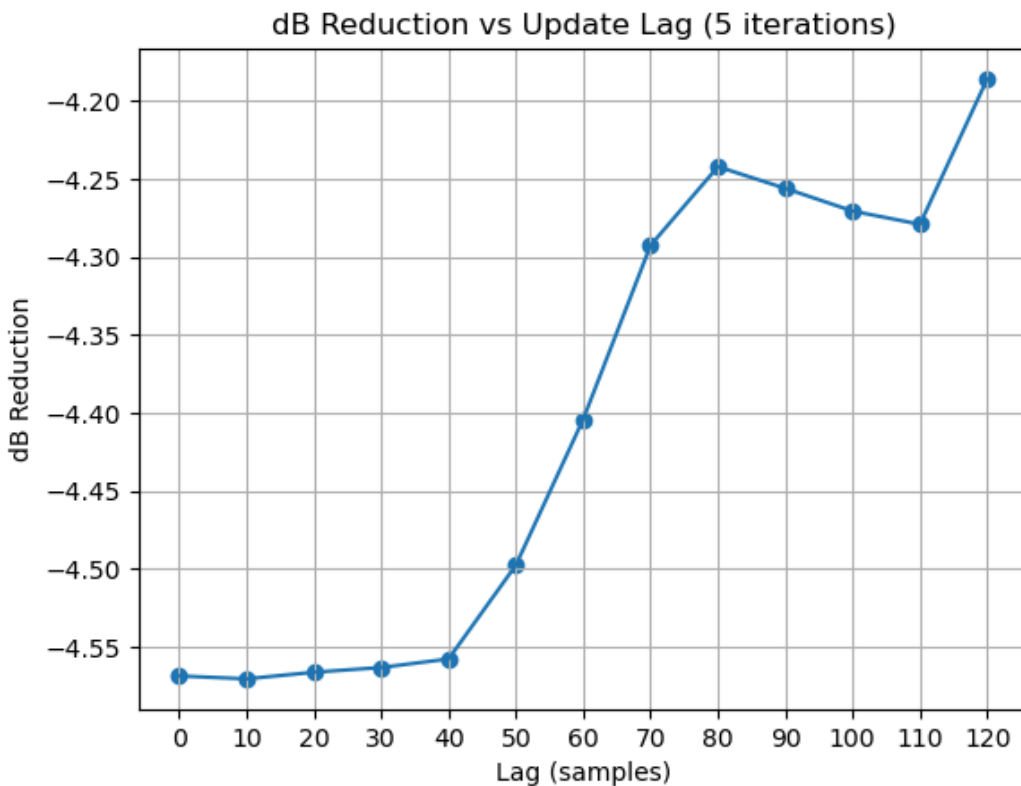


It had to be something with the gradient update

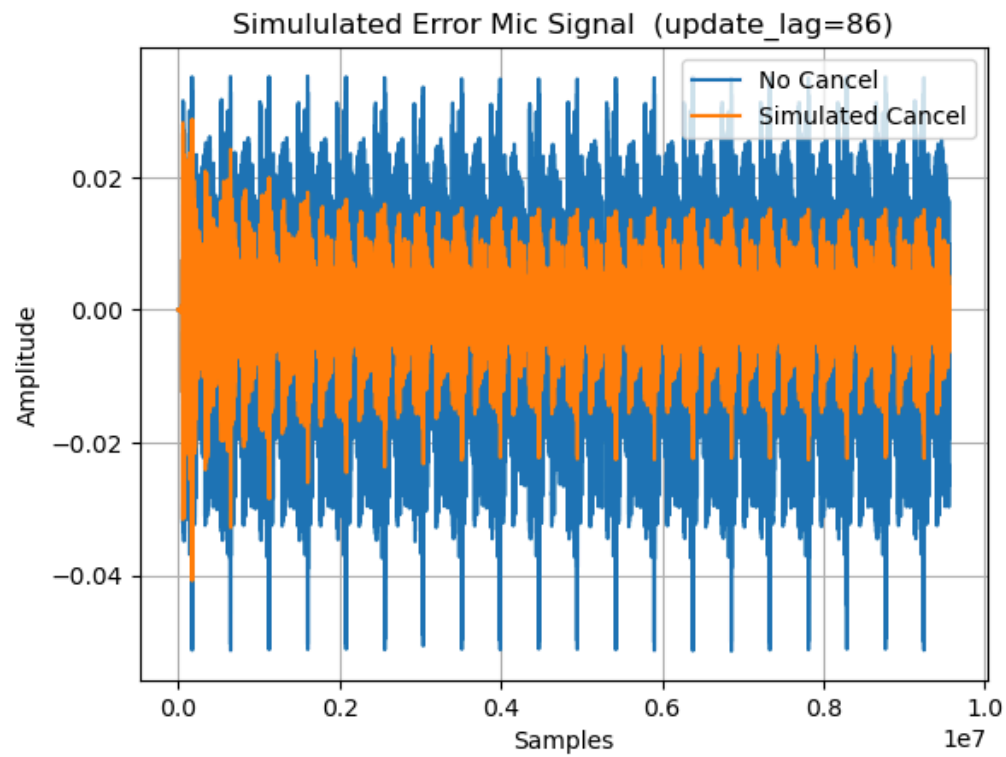
- The actual algorithm was correct, so I verified that the filtered x buffer was correct
- The buffer was only as long as the filter, no extra room for lagged samples
- When we lagged it, the oldest <lag> samples overflowed into the new samples



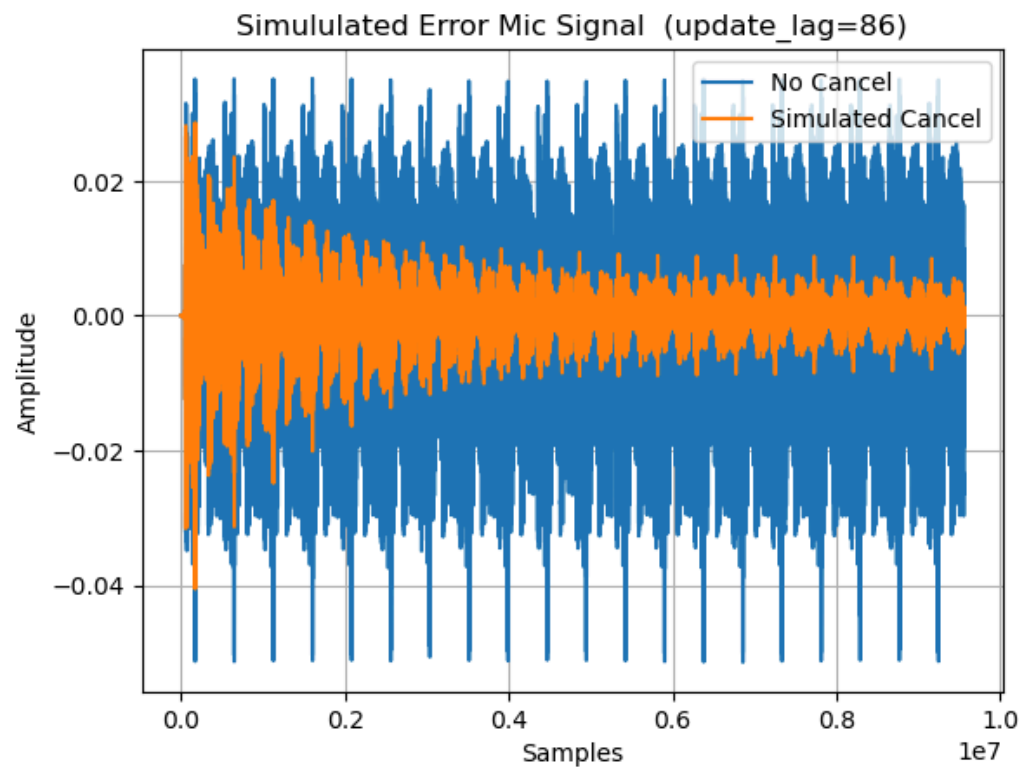
Fixing that fixed the problem



With original params -9.58dB



With no leak, -12.62dB



- It looks like it was just a typo carried over to the SHARC
- This error didn't exist on the code for Python/Focusrite
- I think the excessive lag on the Focusrite caused the issue in that case